

Assembler, LED, and Text-Based Tamagotchi

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1 Assembler Implementation

Our assembler is composed of three main components: the emulate file, where instruction fetch and encode is performed; the modules, which consist of all instruction-encoding functions; and the data structures. One such data structure is the dynamic **symtable**, a dictionary of labels to addresses. The second data structure is the alias table, a dictionary of mnemonics to function pointers where the instruction type is handled.

This assembler utilizes two-pass assembly. The function **first_pass** loops through the input assembly file, identifying and storing labels in a symbol table with their corresponding addresses. Next, the function **second_pass** loops through the assembly file again, tokenising each line to identify the instruction name, replace labels with offsets, and extract instruction parameters. The instruction name is then inputted to a lookup table to obtain a function pointer to the proper encode function. For example, inputting the **add** instruction to lookup table returns a pointer to function **arith** in module **data-processing**. Each of these functions takes the list of tokens, the number of tokens in the list, and an integer pointer where the encoded instruction is to be stored. The function returns a status indicating encoding success, either **EXIT_SUCCESS** or **EXIT_FAILURE**. Lastly, the encoded instruction is converted to binary and written to the specified output file. Note that in the second pass, any line consisting of only a label, directive, or newline is automatically skipped.

2 Blinking LED Implementation

3 Text-Based Tamagotchi

3.1 Tamagotchi Design

3.2 Challenges

3.3 Testing

4 Group Reflection

5 Individual Reflections

5.1 Tangina Kader

5.2 Amy Showell

5.3 Shriya Sistla

5.4 Yao Tong